

TIC 139805341

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	2.931491 d	VSX	no match
Epoch (t0)	1327.0564 BTJD	ASAS-SN	no match
Duration	0.0840 d	Gaia eclipsing-binary table	no match
Depth	1.083 % (10,834 ppm)	Gaia variable classification	no match
SNR	22.6	TESS EB catalog (Prsa+2022)	no match
		ExoFOP TOI list	no match
INDEPENDENT CHECKS (f3i gates)		Verdict: NOVEL	
Eclipses observed	4 (gate >= 3)		
Robust depth ratio	0.97 (gate >= 0.50)		
Robust depth sigma	20.4 (gate >= 7.0)		
Odd/even z-score	2.88 (consistent)		
Secondary eclipse	not detected		

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 1 light-curve dip in this sector: 19.8 sigma t0 refined by -0.0025 phase

Difference-image peak SNR 5.29 (gate >= 5.0) Compactness 1.00 (gate >= 0.50)

Dimming centroid offset 0.25 px (5 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 139805343

0.50 px away, dTmag +3.85 - farther from the dimming than the target

